

Muhammad Aqeel

POSTDOCTORAL RESEARCH FELLOW · COMPUTER VISION & ANOMALY DETECTION

Dept. of Engineering for Innovation Medicine, University of Verona, Italy

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Research Interests

My research is centered on anomaly detection for industrial inspection, with a focus on unsupervised and zero-shot settings where labeled defect data is not available. Over time my work has naturally expanded toward generative models and vision-language approaches, driven by the need to handle data scarcity and improve generalization across unseen defect categories. Recently I am also exploring video understanding and multimodal summarization, though my main research remains in anomaly detection.

Employment

University of Verona

POSTDOCTORAL RESEARCH FELLOW

Verona, Italy

2026 – Present

University of Ljubljana

VISITING RESEARCHER

Ljubljana, Slovenia

Jun. – Nov. 2025

University of Verona

TEACHING ASSISTANT

Verona, Italy

2023 – 2025

Huazhong Univ. of Science & Technology

RESEARCH ASSISTANT

Wuhan, China

2021 – 2022

Education

University of Verona

PH.D. IN COMPUTER SCIENCE

Supervisors: Prof. Francesco Setti & Prof. Marco Cristani

Verona, Italy

2022 – 2025

Beijing Univ. of Posts & Telecommunications

M.Sc. IN COMPUTER SCIENCE

Supervisor: Prof. Xiaohong Liu

Beijing, China

2019 – 2021

PMAS Arid Agriculture University

B.Sc. IN COMPUTER SCIENCE

Rawalpindi,

Pakistan

2013 – 2017

Publications

JOURNAL ARTICLES

Semantic Parsing for Aspect Based Sentiment Analysis

M. AQEEL & F. SETTI

IEEE Access, Vol. 13, pp. 127322–127334

2025

CONFERENCE PROCEEDINGS

Anomaly-Aware Vision-Language Adapters for Zero-Shot Anomaly Detection

M. AQEEL, M. NAZIR, U. KHAN, M. CRISTANI & F. SETTI

IEEE Intl. Conf. on Image Processing (ICIP)

2026

Multimodal Abstractive Summarization of Instructional Videos with Vision-Language Models

M. NAZIR, R. ZHANG, M. AQEEL & F. SETTI

Intl. Conf. on Pattern Recognition (ICPR)

2026

Towards Real Unsupervised Anomaly Detection via Confident Meta-Learning

M. AQEEL, S. SHARIFI, M. CRISTANI & F. SETTI

IEEE/CVF Intl. Conf. on Computer Vision (ICCV)

2025

Robust Anomaly Detection in Industrial Environments via Meta-Learning

M. AQEEL, S. SHARIFI, M. CRISTANI & F. SETTI

IEEE/CVF Intl. Conf. on Computer Vision (ICCV)

2025

A Contrastive Learning-Guided Confident Meta-Learning for Zero-Shot Anomaly Detection

M. AQEEL, D. SKOČAJ, M. CRISTANI & F. SETTI

IEEE/CVF Intl. Conf. on Computer Vision (ICCV)

2025

ExDD: Explicit Dual Distribution Learning for Surface Defect Detection via Diffusion Synthesis

M. AQEEL, F. LEONARDI & F. SETTI

ICIAP 2025, Springer

2025

RoadFusion: Latent Diffusion Model for Pavement Defect Detection

M. AQEEL, K. D. BELLETE & F. SETTI

ICIAP 2025, Springer

2025

Latent Space Synergy: Text-Guided Data Augmentation for Direct Diffusion Biomedical Segmentation

M. AQEEL, M. NAZIR, Z. RUAN & F. SETTI

ICIAP Workshops 2025, Springer

2025

Diffusion-Based Data Augmentation for Medical Image Segmentation

M. NAZIR, M. AQEEL & F. SETTI

IEEE/CVF Intl. Conf. on Computer Vision (ICCV)

2025

Self-Supervised Iterative Refinement for Anomaly Detection in Industrial Quality Control

M. AQEEL, S. SHARIFI, M. CRISTANI & F. SETTI

VISAPP 2025, SCITEPRESS, pp. 173–183

2025

Meta Learning-Driven Iterative Refinement for Robust Anomaly Detection in Industrial Inspection

M. AQEEL, S. SHARIFI, M. CRISTANI & F. SETTI

ECCV 2024, Springer, pp. 445–460

2024

Self-supervised Learning for Robust Surface Defect Detection

M. AQEEL, S. SHARIFI, M. CRISTANI & F. SETTI

DeLTA 2024, Springer, pp. 164–177

2024

Diffusion-Based Image Generation for In-Distribution Data Augmentation in Surface Defect Detection

L. CAPOGROSSO, F. GIRELLA, F. TAIOLI, M. D. CHIARA, M. AQEEL, F. FUMMI, F. SETTI & M. CRISTANI

VISAPP 2024, SCITEPRESS, pp. 409–416

2024

Funding & Grants

University of Verona

PNRR FUNDING FOR PHD

Verona, Italy

2022 – 2025

University of Ljubljana

MOBILITY RESEARCH GRANT

Ljubljana, Slovenia

2025

Beijing Univ. of Posts & Telecommunications

CHINESE GOVERNMENT SCHOLARSHIP (CSC) FOR MASTER

Beijing, China

2019 – 2021

Awards & Honours

Workshop on CV and Generative Models for Medical Imaging, ICIAP 2025

CERTIFICATE OF ORAL PRESENTATION

Rome, Italy

2025

MUHAMMAD AQEEL · CURRICULUM VITAE2

Academic Service

REVIEWER – JOURNALS

IEEE Transactions on Multimedia

REVIEWER

Neural Computing and Applications

REVIEWER

Applied Soft Computing

REVIEWER

REVIEWER – CONFERENCES

IEEE/CVF International Conference on Computer Vision (ICCV)

REVIEWER

2025

European Conference on Computer Vision (ECCV)

REVIEWER

IJCAI-ECAI

REVIEWER

Intl. Conf. on Computer Vision Theory and Applications (VISAPP)

REVIEWER

2025

Conferences & Presentations

ICIAP 2025

ORAL, POSTER & INVITED TALK

- International Conference on Image Analysis and Processing · Rome, Italy · Sep. 2025
- Oral: *RoadFusion: Latent Diffusion Model for Pavement Defect Detection*
- Poster: *ExDD: Explicit Dual Distribution Learning for Surface Defect Detection*
- Oral: *Latent Space Synergy* – CVGMMI Workshop
- Invited Talk – T4S Workshop (Traffic & Street Surveillance for Safety and Security)

ECCV 2024

ORAL PRESENTER

- European Conference on Computer Vision · Milan, Italy · Sep.–Oct. 2024
- Oral: *Meta Learning-Driven Iterative Refinement for Robust Anomaly Detection* – VISION Workshop

VISAPP 2025

ORAL PRESENTER

- Intl. Conf. on Computer Vision Theory and Applications · Porto, Portugal · Feb.–Mar. 2025
- Oral: *Self-Supervised Iterative Refinement for Anomaly Detection in Industrial Quality Control*

VISAPP 2024

POSTER PRESENTER

- Intl. Conf. on Computer Vision Theory and Applications · Rome, Italy · Feb. 2024
- Poster: *Diffusion-based Image Generation for In-distribution Data Augmentation in Surface Defect Detection*

Memberships

CVPL – Italian Association for Computer Vision, Pattern Recognition and Machine Learning

MEMBER

Italy

Since 2025

Teaching Assistantship

- Computer Vision and Machine Learning, Dept. of Engineering for Innovation Medicine

Technical Skills

Languages	Python, C, C++, Java, MATLAB
Frameworks	PyTorch, TensorFlow, Keras, Hugging Face
Tools	Git, Docker, ￼, Anaconda, PyCharm, Visual Studio
Databases	MySQL, Oracle, Neo4j
Operating Systems	Linux, macOS, Windows

Languages

Urdu / Punjabi	Native
English	Proficient (C2)
Italian	Basic (A2)
Chinese	Basic (A2)

Referees

Prof. Francesco Setti

ASSOCIATE PROFESSOR

Dept. of Engineering for Innovation Medicine, University of Verona · francesco.setti@univr.it

Verona, Italy

Prof. Marco Cristani

PROFESSOR

Dept. of Engineering for Innovation Medicine, University of Verona · marco.cristani@univr.it

Verona, Italy

Prof. Liu Xiao Hong

ASSOCIATE PROFESSOR

School of Computer Science, Beijing Univ. of Posts & Telecommunications · xiaohongliu@bupt.edu.cn

Beijing, China